

Roll No.

TCS-372

**B. TECH. (CSE) (THIRD SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022
MATHEMATICS FOR AI AND MACHINE LEARNING**

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Compare and contrast all the level of measurement for data with suitable examples. (CO1)

OR

- (b) In the context of data analysis, what is the difference between parameter and statistics ? (Explain with formulae, if required.) (CO1)
2. (a) Number of days during which nine members of staff did not use their cars : (CO2)

2 6 2 4 1 4 3 1 1

- (i) Determine mean, median and mode for above.
(ii) Determine the interquartile range.

OR

- (b) As we know that data can be categorized in the form of structured, semi-structure and unstructured, elaborate the data and its taxonomies in the context of big data with its use cases. (CO2)

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(2)

3. (a) Find the population variance of the age of children in a family of five children aged 16, 11, 9, 8 and 1. Explain the condition where we apply central limit theorem. (CO1)

OR

- (b) Take an example of any social media platform (such as : facebook) and explain variety, volume and velocity of data generation in Big data. (CO1)

4. (a) Explain the difference between conditional probability and independence. What are Union and Intersection in probability ? (CO2)

OR

- (b) In a supermarket if the mean of expenditure of a shopper is 50 Euros, standard deviation is 15 and $n = 500$. Find the value below which 70% of shoppers are expected to spend. ($Q = 0.2000 \Rightarrow z = 0.524$). (CO2)

5. (a) Explain and visualize the rectangular and bimodal distribution in continuous distribution. (CO3)

OR

- (b) What is normal distribution ? Also explain the different shapes of probability distribution in the context of skewness. (CO3)